Dummy Variables for Two Levels

Class	Levels	Dummy Variable 1
Decision	Innocent	1
	Guilty	0

Implementation:

Row wise

- For k-levels in a categorical variable we have k-1 dummy variables
- For 2-levels of categorical variable "Decision" we have **1 dummy variable** to represent either of the two levels
- This dummy variable may take 1 for innocent or 0 for guilty
- If you use both 0 and 1 i.e., [0 1] for innocent and [1 0] for guilty, you end up with perfect collinearity and dummy variable trap

Dummy Variables (DV) for Three Levels

Class	Levels	Dummy Variable 1	Dummy Variable 2
Colour	Green	1	0
	Orange	**0**	**1**
	Red	0	0

Implementation:

Row wise

- For k-levels in a categorical variable we have k-1 dummy variables
- For 3-levels of categorical variable "Colour" we have **2 dummy variables** to represent either of the three levels

Dummy Variables (DV) for Four Levels

Class	Levels	Dummy Variable 1	Dummy Variable 2	Dummy Variable 3
Weather	Winter	0	0	0
	Spring	1	0	0
	Summer	0	1	0
	Fall	0	0	1

Implementation:

Row wise

- For k-levels in a categorical variable we have k-1 dummy variables
- For 4-levels of categorical variable “Weather” we have **3 dummy variables** to represent each of the four levels

Dummy Variables (DV) for Four Levels

Class	Levels	Dummy Variable 1 Is Spring	Dummy Variable 2 Is Summer	Dummy Variable 3 Is Fall
Weather	Winter	0	0	0
	Spring	**1**	**0**	**0**
	Summer	0	1	0
	Fall	0	0	1

Implementation:

Row wise

3. The Truth Table

Here is how your dataset will actually look for each season:

Actual Season	`Is_Spring`	`Is_Summer`	`Is_Fall`
Winter (Baseline)	0	0	0
Spring	1	0	0
Summer	0	1	0
Fall	0	0	1

Collinearity and Multicollinearity

- Collinearity occurs when exactly two independent variables have a linear relationship. If one variable changes, the other changes in a predictable, linear fashion.
- Multicollinearity occurs when three or more independent variables are correlated, or
- when one independent variable is a linear combination of two or more other independent variables.

Linear Regression with OHE Variables

- Let there be a mix of categorical and numerical predictors.
- Using 1-hot encoding all columns (of 1s and 0s) are present
- To avoid multicollinearity we do not include the intercept in the model

A. So, what does the final regression equation represent?

B. How do you use the equation as a predictor for a new setting of numerical and/or categorical variable?

Linear Regression with OHE Variables

A. So, what does the final regression equation represent?

B. How do you use the equation as a predictor for a new setting of numerical and/or categorical variable?

Linear Regression with OHE Variables

A. So, what does the final regression equation represent?

Assume that a categorical variable has

$$\widehat{\mathrm{Y}} = \widehat{\beta}_1 \mathrm{X}_1 + \widehat{\beta}_2 \mathrm{X}_2 + \widehat{\beta}_3 \mathrm{X}_3 + \widehat{\beta}_4 \mathrm{X}_4$$

More meaningfully, House Price depends on square feet and City

$$\widehat{\mathrm{P}} = \widehat{\beta}_1 \mathrm{SquareFeet} + \widehat{\beta}_2 \mathrm{Chennai} + \widehat{\beta}_3 \mathrm{Mumbai} + \widehat{\beta}_4 \mathrm{Calcutta}$$

Note one-hot encoding used to represent cities and hence no intercept

is used (to avoid multicollinearity)

Regression with OHE of Categorical Variables

$$\hat{Y} = \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3 + \hat{\beta}_4 X_4$$

More meaningfully, House Price depends on square feet and City

$$\hat{P} = \hat{\beta}_1 \mathrm{SquareFeet} + \hat{\beta}_2 \mathrm{Chennai} + \hat{\beta}_3 \mathrm{Mumbai} + \hat{\beta}_4 \mathrm{Calcutta}$$

One-hot encoding represents cities & no intercept is used (avoid multicollinearity)

Class	Levels	One-Hot encoding
City	Chennai	1 0 0
	Mumbai	**0 1 0**
	Calcutta	0 0 1

Regression with OHE Variables: Intercept Removed

$$\hat{P} = \hat{\beta}_1 \text{SquareFeet} + \hat{\beta}_2 \text{Chennai} + \hat{\beta}_3 \text{Mumbai} + \hat{\beta}_4 \text{Calcutta}$$

Class	Levels	Dummy Variable encoding	Square Feet
City	Chennai	1 0 0	2000
	Mumbai	**0 1 0**	**1000**
	Calcutta	0 0 1	1500

Class	Levels	Square Feet	Predicted $\hat{P}$
City	Chennai	2000	$\hat{\beta}_1\{2000\} + \hat{\beta}_2$
	Mumbai	**1000**	$\hat{\beta}_1\{1000\} + \hat{\beta}_3$
	Calcutta	1500	$\hat{\beta}_1\{1500\} + \hat{\beta}_4$

Even though you removed intercept, we end up with a new intercept

Why Drop Intercept? Drop one Column in OHE

Is it possible to retain intercept but drop the first column in OHE variables to avoid multicollinearity?

Class	Levels	Dummy Variables	Square Feet
City	Chennai	~~1~~ 0 0	2000
	Mumbai	~~0~~ **1 0**	**1000**
	Calcutta	~~0~~ 0 1	3500

- Yes, it becomes dummy variables encoding, with only two dummy variables for 3 cities, and Chennai becoming reference with [0 0]

Linear Regression with Dummy Variables

$$\hat{P} = \hat{\beta}_0 + \hat{\beta}_1 \text{SquareFeet} + \cancel{\hat{\beta}_2 \text{Chennai}} + \hat{\beta}_2 \text{Mumbai} + \hat{\beta}_3 \text{Calcutta}$$

Class	Levels	Dummy Variable encoding	Square Feet
City	Chennai	0 0	2000
	Mumbai	**1 0**	**1000**
	Calcutta	0 1	1500

Class	Levels	Square Feet	Predicted $\hat{P}$
City	Chennai	2000	$\hat{\beta}_0 + \hat{\beta}_1\{2000\}$
	Mumbai	**1000**	$\hat{\beta}_0 + \hat{\beta}_1\{1000\}+\hat{\beta}_2$
	Calcutta	1500	$\hat{\beta}_0 + \hat{\beta}_1\{1500\} + \hat{\beta}_3$

Even though you removed 1 OHE column, we may end up with two intercepts